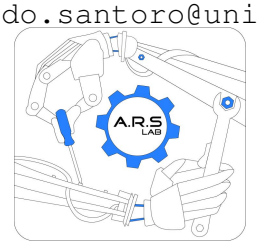
Predictive Filters

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Advanced Robotics and Autonomous Systems

Notes

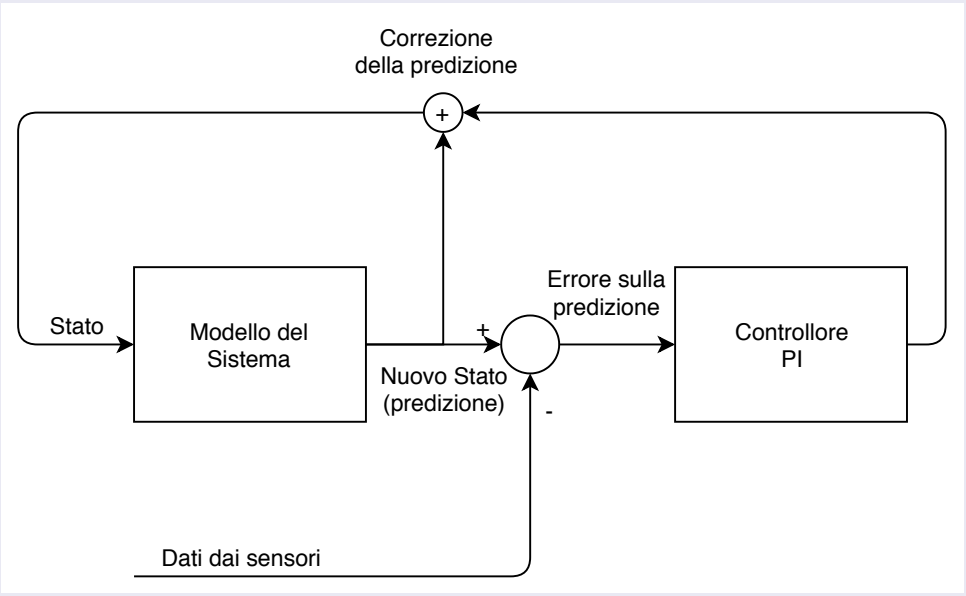
Predictive Filters

Predictive Filters

- **Predictive Filters** are algorithms able to predict the state of an evolving system
- A Predictive Filter needs:
  - A **model** of the system
  - Some **sensor** able to acquire the actual values of one or more state variables
- By using the model, next values of state variables is **estimated**
- The state estimate is thus **compared** against sensor values and an **error** is computed
- The error is given to a **PI controller** and its output is **summed** to the state variables in order to adjust the estimate

Notes

Predictive Filter



Notes

Predictive Filters

Predictive Filters Tuning

- Predictive filters can reduce problems related to measurement noise and offset
- But pose the problem of **tuning** of constants K_P and K_I
- **"Small values"** produce low error correction
- **"High values"** improve error correction but cause overshots and oscillations in the estimate

Notes

Kalman Filter

Kalman Filter

- The **Kalman Filter** is a predictive filter able to solve the problem of the tuning of constants
- It is based on the same principle of the LQR control since it has the objective of **minimizing the estimate error**
- Indeed, it computes an **optimal gain** K, but...
- The gain is updated (recomputed) for each cycle
- Optimality is based on a statistical approach assuming that the statistics of model (process) noise and measurement noise are known
- It is often referred as **LQG, Linear Quadratic Gaussian** filter

Notes

Kalman Filter

System Model

The (discrete) system is characterized by the update law:

$$\mathbf{x}(k+1) \quad = \quad A\,\mathbf{x}(k) + \mathbf{w}$$

wehere:

- $\mathbf{x}$: state vector
- A: state matrix
- $\mathbf{w}$: uncertainty (noise) vector of the state variables

$\mathbf{w}$ is a vector of normal (gaussian) random variables with zero mean and variance known

Notes

Measure Model

Let's assume we have sensors that are able to measure a subset of state variables

Let's call $\mathbf{z}(k)$ the measure vector at discrete instant k, so we can model the measure process as:

$$\mathbf{z}(k) = H\,\mathbf{x}(k) + \mathbf{v}$$

where:

- H: measure matrix, it specifies what variables are measured
- $\mathbf{v}$: uncertainty (noise) vector of the measures

$\mathbf{v}$ is a vector of normal (gaussian) random variables with zero mean and variance known

Notes

Kalman Filter

Statistics

- **w** and **v** are vector of random variables
- Kalman filter needs to know the statistics of these random variables
- In particular, they must be **zero-mean gaussian** with variance and co-variance known
- For each of these vector, a **covariance matrix** must be defined

Notes

Covariance Matrix

Let's assume $\mathbf{s}$ a vector of random variables, where $\bar{\mathbf{s}}$ is the vector of average values computed over a set of N samples $\mathbf{s}_i$

The **covariance matrix** is defined as

$$\Sigma = \frac{1}{N-1} \sum_{i=1}^{N} (\mathbf{s}_i - \bar{\mathbf{s}})(\mathbf{s}_i - \bar{\mathbf{s}})^T$$

If the single variables of the vector are sono statistically independent Σ is a **diagonal matrix** and the elements of the diagonal are the variances of each variable

Notes

Covariance Matrices

On the basis of the statistical behaviour of **w** and **v** the following matrices are defined:

- Q: covariance matrix of **w** (model uncertainty)
- R: covariance matrix of **v** (measure uncertainty)

The Kalman filter algorithm employs another matrix, called P which is the **covariance matrix of the error** between the estimated state and the measured state

P is initially set-up to a initialization value P_0 and is updated for each step of the algorithm using the system model

Often P_0 is initialized with the identity $P_0 = I$

Notes

Algorithm

1. **PREDICTION**
   - A new estimate of the state vector is derived from system model
   - The error covariance matrix is updated according to system model
2. **UPDATE**
   - A new measure is obtained from sensors and the error is computed
   - An optimal gain is computed and it is used, with the error, to correct the estimate
   - The error covariance matrix is also updated according to error and the optimal gain

Notes

Kalman Filter

Algorithm

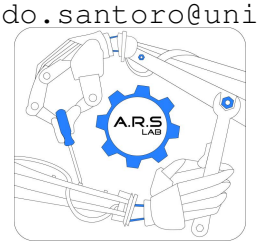
1. **PREDICTION**
   - Estimate update: $\mathbf{x}_k = A\,\mathbf{x}_{k-1}$
   - Error Covariance update: $P_k = A\,P_{k-1}\,A^T + Q$
2. **UPDATE**
   - New measure: $\mathbf{z}_k$
   - Optimal gain: $K_k = P_k\,H^T\,(H\,P_k\,H^T + R)^{-1}$
   - State correction: $\mathbf{x}_k = \mathbf{x}_k + K(\mathbf{z}_k - H\mathbf{x}_k)$
   - Error Covariance matrix correction: $P_k = (I - K_k\,H)\,P_k$

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